DocGuard

Document tampering localization
CNN and Transformer comparison

Deep Learning and GenAI | CS312B
B.Tech CS | Semester 5 | Section D

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The problem

A page label does not explain where an edit occurred.

Goal: predict suspicious regions and make them reviewable.

Scope: controlled research on synthetic receipts.
Real document detection accuracy is not established.

From document to review

Upload image or PDF → select page

Normalize and resize → CNN or Transformer

Pixel scores → fixed validation threshold

Overlay and heatmap → PNG, PDF and ZIP exports

A controlled synthetic dataset

600 fictional receipt sources → 1,200 images

840 train / 180 validation / 180 test
Source identities and layout configurations are disjoint.

Authentic and edited images share acquisition variation.
Masks label edited rectangles, not individual characters.

Two segmentation approaches

Compact CNN
30,081 parameters; local feature extraction and skip connections.

MiT B0 Transformer plus custom multiscale decoder
3,349,377 parameters; ImageNet pretrained encoder.

Compared frozen encoder and full fine tuning. No GAN trained.

Training and fair evaluation

Three CPU epochs | 160 × 200 input | batch 8 | seed 42

Binary cross entropy plus positive document Dice loss.
Checkpoint and thresholds selected on validation data.

Global pixel precision, recall, F1 and IoU on 180 test images.
Planned longer Colab training has not been executed.

Actual test results

Model	Precision	Recall	F1	IoU
CNN	0.1878	0.3565	0.2460	0.1403
Frozen Transformer	0.1330	0.6283	0.2195	0.1233
Fine tuned Transformer	0.1691	0.3538	0.2288	0.1292

The Transformer did not outperform CNN global pixel F1.

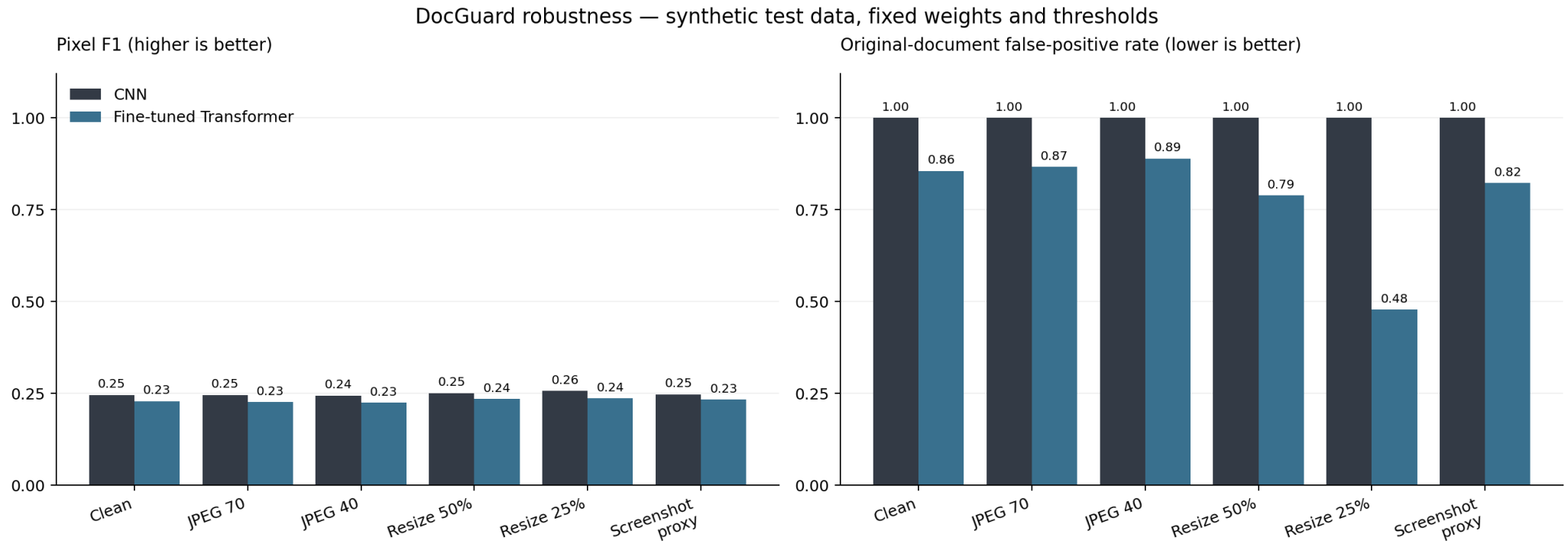
False positives are the main limitation

Authentic test pages incorrectly flagged

CNN: 90 / 90 · 100%
Fine tuned Transformer: 77 / 90 · 85.6%

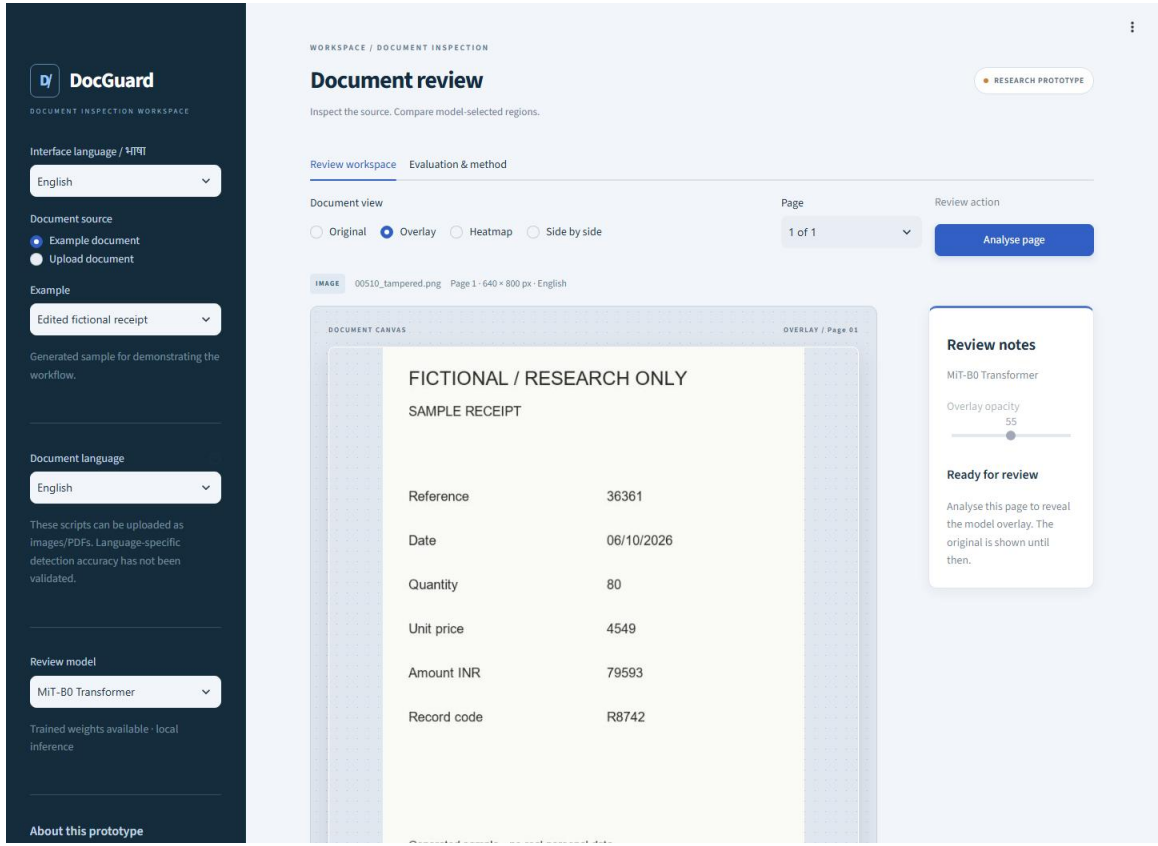
Flag rule: at least 0.1% of model pixels predicted positive.
Highlighted area is not a fraud probability.

Robustness under acquisition changes



JPEG, down and up resizing, and simulated screenshot acquisition.

A working local review application



Regional language workflow

Interface: English, Hindi, Marathi, Tamil and Bengali.

Hindi PDF and regional language examples are included.
Language switches preserve the current review.

No OCR or translation is performed.
Language specific detection accuracy remains unvalidated.

Five minute demonstration

1. Analyse an edited English sample using CNN.
2. Compare the Transformer overlay.
3. Upload the supplied Hindi PDF.
4. Switch interface language and export the review.
5. Explain an authentic page false positive.

Verification and reproducibility

14 unit checks passed.
14 browser flows and 7 language checks passed.

Source, trained weights, splits and metrics are supplied.
CPU model forward medians: CNN 19.70 ms; Transformer 34.65 ms.
These timings exclude application and PDF processing.

What the project establishes

A reproducible CNN and Transformer experiment.
A complete local document review workflow.
A clear account of weak detection performance.

Next: licensed real documents, higher resolution, longer training and independent calibration.